

James Nergard

(602) 799-9441 • nergardj@my.erau.edu • <https://jnergard.github.io/JNergard-Website>

Education

Bachelor of Science, Software Engineering
Embry-Riddle Aeronautical University (ERAU) - Prescott, Arizona

GPA: 4.0/4.0
05/2028

Skills

Software Development & Testing: Software Testing, Quality Assurance, Test-Driven Development, Agile Project Management, Microprocessor Coding, GitHub, Docker, VSCode, Object-Oriented Programming
Programming Languages: Python, C++, C, Arduino, ARM Assembly, SQL
Engineering Tools: Microsoft Office, Autodesk Fusion 360, SolidWorks

Project Experience

Competition Robotics – FIRST Robotics

08/2020 – 07/2024

Head of Coding

- Used GitHub and Visual Studio Code for version control and software development
- Programmed the entire robot in C++ with guidance from my mentor
- Awarded the 2021 Creativity Award for sophisticated code and innovative 3D prints
- Designed the entire robot in Fusion 360 (CAD) and 3D printed parts
- Robot won the 2024 ASU State Finals

File Deletion System

01/2026 – 05/2026

- Created an application to delete excess Windows files
- Coded in Python
- Link to GitHub: <https://github.com/JNergard>

Research Paper on Partial Differential Equations

12/2024 – 05/2025

- Worked on a research paper with a professor and colleague
- Solved renowned PDEs, such as the heat and Schrödinger equations
- DOI: 10.30919/es1573

Golf Bracket Tournament Tracker

08/2026 – 09/2026

- Built a Python engine for a Swiss-style, 20-player bracket with 2-loss elimination
- Designed a Buchholz resistance algorithm to rank and fold-pair players, avoiding rematches
- Developed cascading “pairing-down” logic to resolve uneven brackets each round
- Deployed a Flask web app using the GitHub API for persistence on free, ephemeral hosting
- Wrote a 23-test pytest suite covering the player, pairing, and tournament logic

Professional Experience

InRange Golf

Remote

Software Testing

07/2026 – 12/2026

- Tested a new software build for InRange Golf
- Developed test cases and applied software testing principles
- Prepared and submitted a report documenting findings, fixes, and suggestions

Dobson Ranch Golf Course

Mesa, AZ

Outside Services

05/2021 – 05/2022

- Provided customer service for all patrons
- Communicated over radio with a team of over 20+ people simultaneously
- Operated and maintained golf carts, other vehicles, and heavy equipment

Quantum Golf Program

Mesa, AZ

Assistant Coach

05/2022 – 07/2026

- Coordinated with different courses to attain different places for students to practice at
- Crafted various lesson plans for the students
- Helped design and create an informative and functional website

Athletics

Embry-Riddle Men's Golf Team

08/2024 – 05/2028

- Commit 25+ hours/week for practice, competition, and travel
- Maintained full-time coursework while also doing team responsibilities
- NAIA and GSAC Scholar Athlete Award recipient